Physics-Inspired Joint-Embedding Predictive Architecture (JEPA) Triad

Representation-space latent predictive dynamics constrained by fundamental physical conservation invariants

(A) Observation Drowning vs. JEPA

Observation-Space Baselines

TimesNet / TranAD / AnomalyTransf.

$\min_{\theta} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$ in raw sensor domain

- Forces networks to fit high-frequency stochastic sensor noise & benign jitter.
- High reconstruction residual $\neq$ true anomaly.
- Generates $>1{,}000{,}000$ false alarms in telemetry.

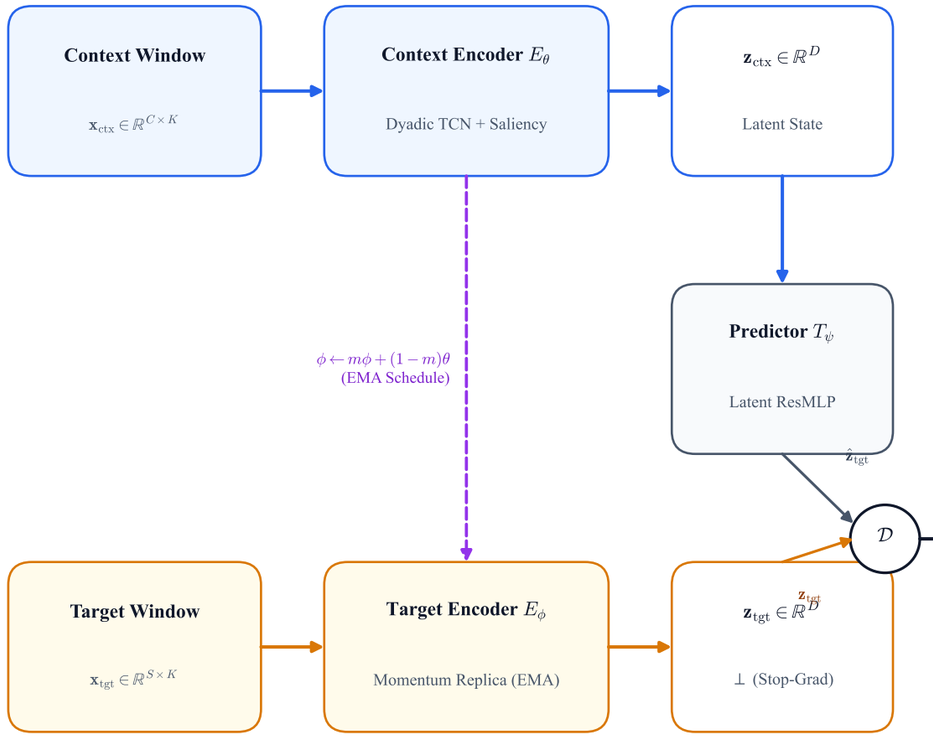
Proposed Latent JEPA Triad

Operator Entropy / Reynolds / Potential Flow

$\min_{\theta,\psi} \mathcal{D}(T_\psi(\mathbf{z}_{\text{ctx}}), \mathbf{z}_{\text{tgt}})$ in $\mathbb{R}^D$

- Predicts abstract invariant dynamical state.
- Discards pixel/point stochastic noise.
- Immune to representation collapse via non-contrastive physical conservation heads.

(B) Unified Predictive Latent Architecture



(C) Physics Heads & Tail Calibration

1. Operator-Entropy JEPA

Koopman Dynamic Spectral Entropy (Unitary Spectrum)

$\mathcal{K}_{\mathbf{z}} \mathbf{z}_t \approx \mathbf{z}_{t+1}, \quad \mathcal{H}_K = -\sum_i \lambda_i \ln \lambda_i$

2. Reynolds-Stress JEPA

Turbulent Momentum Transfer & Shear Divergence

$\mathbf{u} = \bar{\mathbf{u}} + \mathbf{u}', \quad \mathbf{R} = \overline{\mathbf{u}' \otimes \mathbf{u}'} \quad (\nabla \cdot \mathbf{R} \neq 0)$

3. Potential-Flow JEPA

Conservative Irrotational Potential Field

$\mathbf{v} = \nabla \Phi, \quad \nabla \times \mathbf{v} = 0 \quad$ (Vorticity-Free Conservation)

Pure Inductive EVT & Streaming SPOT

Kurtosis-adaptive Generalized Pareto distribution tail fitting

$\tau(t) = \text{Med}(t) + (\tau_0 - \text{Med}_0) \cdot \frac{\text{IQR}(t)}{\text{IQR}_0}$